# Agentic X-TEC

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**Background:** X-TEC is a machine learning pipeline that researchers use to analyze massive amounts of X-ray diffraction data from quantum materials. It uses unsupervised clustering algorithms to group data points based on how they react to temperature changes, successfully uncovering hidden physical phases. Traditional X-TEC heavily relies heavily on human intervention. Currently, a researcher must specify parameters, like number of clusters, based on trial, error, and guesswork.

**Implementation:** To put this automation into practice, I have built Python code that navigates and reads through NeXus files to pull out X-ray data. Now that the data can be extracted automatically, I am currently implementing and experimenting with different types of unsupervised clustering algorithms within the X-TEC pipeline. By testing and comparing how these clustering methods handle data, I am establishing a benchmark to see which model works best.

**Goals:** The goal this project is to build an agentic AI around the existing X-TEC to completely automate the data analysis. Currently, researchers spend an immense amount of time collecting raw X-ray data and it also takes a long time to process the data before it can be fed into X-TEC. By making X-TEC automated it will determine key parameters without any human direction. This will accelerate the discovery of new quantum materials.

X-ray Temperature Clustering (X-TEC) Pipeline, Kim Group GitHub.